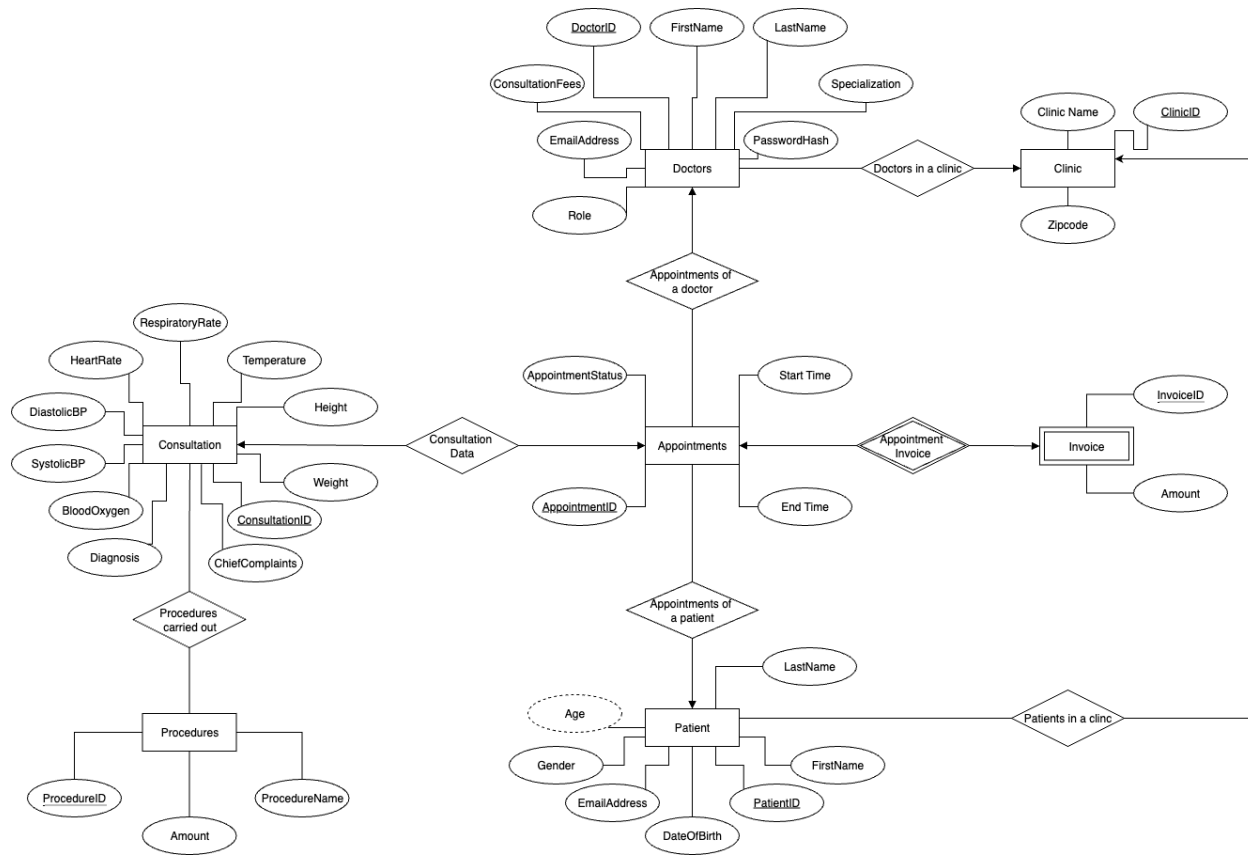


# Digital Clinic Management System

## ER Diagram



## Entities

**Doctors:** The primary users of the platform.

- **DoctorID:** The unique identifier for each doctor on the platform.
- **FirstName:** First name of the doctor.
- **LastName:** Last name of the doctor.
- **EmailAddress:** A unique email address of the doctor, which is used to sign into the platform.
- **PasswordHash:** Hashed password of the doctor's account, which is used to compare with the entered password during sign-in.
- **Specialization:** Specialization of the doctor (general physician, dietician, dentist, etc.)
- **ConsultationFees:** Fixed amount added to the invoice on the completion of each appointment by the doctor.

- Role: Role of the doctor, can be one of admin or consultation. Admin can see all data of all the doctors present in the clinic.

**Clinics:** The entity holding a group of doctors, who have shared access to patients and consultations.

- ClinicID: The unique identifier for each clinic on the platform.
- ClinicName: The name of the clinic.
- Zipcode: The 6 digit zipcode of where the clinic is located.

**Patients:** The patients created by doctors, against which appointments are created

- PatientID: The unique identifier for each patient in each clinic on the platform.
- FirstName: First name of the patient.
- LastName: Last name of the patient.
- Age: Age of the patient, a derived attribute, which is calculated at query runtime using the date of birth.
- DateOfBirth: Date of birth of the patient.
- EmailAddress: Email address of the patient, used to contact the patient.
- Gender: Gender of the patient.

**Appointments:** The appointments created by doctors against patients

- AppointmentID: The unique identifier of each appointment on the platform.
- AppointmentStatus: Status of appointment, one of booked, waiting in clinic, active, and completed.
- StartTime: The scheduled starting time for the appointment.
- EndTime: The scheduled end time for the appointment.

**Consultation:** The data (diagnosis recorded and stored against each appointment conducted by the doctor)

- ConsultationID: The unique identifier for each consultation against an appointment.
- HeartRate: Heart rate in BPM of the patient measured during the appointment.
- RespiratoryRate: Respiratory rate of the patient measured during the appointment.
- Temperature: Body temperature in Fahrenheit of the patient measured during the appointment.
- SystolicBP: Systolic blood pressure of the patient measured during the appointment.
- DiastolicBP: Diastolic blood pressure of the patient measured during the appointment.
- BloodOxygen: Blood oxygen level of the patient measured during the appointment.
- Height: Height of the patient in centimeters measured at the time of appointment.
- Weight: Weight of the patient in kilograms measured at the time of appointment.
- ChiefComplaints: Complaints by the patient as written down by the doctor for the particular appointment.
- Diagnosis: Diagnosis by the doctor is written down for the particular appointment.

**Procedures:** The list of all medical procedures offered by all the clinics.

- ProcedureID: The unique identifier of each procedure available.
- Procedure Name: A unique medical name for the procedure.

- Amount: The cost of the procedure.

**Invoice:** The invoice against each completed appointment. An appointment is necessary for an invoice to be present, making it a weak entity. It is modeled as a separate entity to allow room for expansion into a detailed list of invoice items, standalone invoices, and separation of concerns from appointment-related data.

- InvoiceID: The unique identifier of the invoice for an appointment.
- Amount: The total charges to be paid for the appointment, calculated on appointment completion, from the consultation fees of the doctor and all procedures performed during the consultation.

## Relationships

**Doctors in a clinic:** Doctors mapped against the clinic that they belong to. Each clinic can have multiple doctors, and one doctor can only be a part of one clinic. **(one to many)**

**Appointments of a doctor:** Appointments mapped against the doctor who conducted them. Each doctor can have multiple appointments, and each appointment can only have one doctor associated with it. **(many to one)**

**Patients in a clinic:** Patients mapped against the clinic that they were created at. Each clinic can have multiple patients, and one patient can only be a part of one clinic. If the same patient goes to two clinics, then a new record is created for the new clinic, keeping patient data between clinics separate. **(many to one)**

**Appointments of a patient:** Appointments mapped against the patient who underwent them. Each patient can have multiple appointments, and each appointment can only have one patient associated with it. **(many to one)**

**Appointment Invoice:** Invoice and cost associated against each appointment. Each appointment can have one and only one invoice, and each invoice is related to one and only one appointment. This relation is a weak relationship, as it involves the Invoice entity, which is a weak entity. **(one to one)**

**Consultation Data:** Consultation data recorded during each appointment. Each appointment can have one and only one consultation record, and each consultation record is related to one and only one appointment. **(one to one)**

**Procedures carried out:** List of procedures carried out during each consultation. Each consultation can have multiple procedures, and each procedure can be associated with multiple consultations. **(many to many)**

# Relational Schema

**Clinics**(ClinicID: INT [PK], ClinicName: VARCHAR(255), Zipcode: CHAR(6))

**Doctors**(

DoctorID: INT [PK],

FirstName: VARCHAR(100),

LastName: VARCHAR(100),

EmailAddress: VARCHAR(255),

PasswordHash: VARCHAR(255),

Specialization: VARCHAR(100),

ConsultationFees: DECIMAL(10,2),

Role: VARCHAR(10),

ClinicID: INT [FK to Clinics.ClinicID]

)

**Patients**(

PatientID: INT [PK],

FirstName: VARCHAR(100),

LastName: VARCHAR(100),

DateOfBirth: DATE,

Age: INT,

EmailAddress: VARCHAR(255),

Gender: CHAR(1)

)

**Appointments**(

AppointmentID: INT [PK],

AppointmentStatus: CHAR(3),  
StartTime: DATETIME,  
EndTime: DATETIME,  
DoctorID: INT [FK to Doctors.DoctorID],  
PatientID: INT [FK to Patients.PatientID]

)

**Invoices(**

InvoiceID: INT [PK],  
Amount: DECIMAL(10,2),  
AppointmentID: INT [FK to Appointments.AppointmentID]

)

**Consultations(**

ConsultationID: INT [PK],  
HeartRate: INT,  
RespiratoryRate: INT,  
Temperature: DECIMAL(4,1),  
SystolicBP: INT,  
DiastolicBP: INT,  
BloodOxygen: DECIMAL(4,1),  
Height: DECIMAL(6,2),  
Weight: DECIMAL(6,2),  
ChiefComplaints: TEXT,  
Diagnosis: TEXT,  
AppointmentID: INT [FK to Appointments.AppointmentID]

)

**Procedures(**

ProcedureID: INT [PK],

ProcedureName: VARCHAR(255),

Amount: DECIMAL(10,2)

)

**ConsultationProcedure(**

ProcedureID: INT [PK],

ConsultationID: INT [PK],

ConsultationID: INT [FK to Consultations.ConsultationID],

ProcedureID: INT [FK to Procedures.ProcedureID]

)

## Functional Dependencies and BCNF Normalization

**Clinic:**

ClinicID  $\rightarrow$  ClinicName, Zipcode

Here, ClinicID is a candidate/primary key, hence in BCNF.

**Doctors:**

DoctorID  $\rightarrow$  FirstName, LastName, Specialization, ConsultationFees, EmailAddress, PasswordHash, Role, ClinicID

EmailAddress  $\rightarrow$  DoctorID

Here, DoctorID is a candidate/primary key. The EmailAddress is also a candidate key, as it is unique and determines the DoctorID. Both determinants are candidate keys, hence in BCNF.

**Patients:**

PatientID  $\rightarrow$  FirstName, LastName, Gender, DateOfBirth, EmailAddress, ClinicID

Here, PatientID is a candidate/primary key, hence in BCNF.

**Appointments:**

AppointmentID  $\rightarrow$  AppointmentStatus, StartTime, EndTime, DoctorID, PatientID

Here, AppointmentID is a candidate/primary key, hence in BCNF.

**Consultations:**

ConsultationID → HeartRate, DiastolicBP, SystolicBP, RespiratoryRate, Temperature, Height, Weight, BloodOxygen, Diagnosis, ChiefComplaints, AppointmentID  
AppointmentID → ConsultationID

Here, ConsultationID is a candidate/primary key. The AppointmentID is also a candidate key, as it determines the ConsultationID (one-to-one mapping). Both determinants are candidate keys, hence in BCNF.

**Invoices:**

InvoiceID → Amount, AppointmentID  
AppointmentID → InvoiceID

Here, InvoiceID is a candidate/primary key. The AppointmentID is also a candidate key, as it determines the InvoiceID (one-to-one mapping). Both determinants are candidate keys, hence in BCNF.

**Procedures:**

ProcedureID → ProcedureName, Amount  
ProcedureName → ProcedureID

Here, ProcedureID is a candidate/primary key. The ProcedureName is also a candidate key, as it is unique and determines the ProcedureID. Both determinants are candidate keys, hence in BCNF.

**ConsultationProcedure:**

(No FDs)

Since there are no FDs, therefore already in BCNF.